Super – High Energy Replicated Optics (SuperHERO)
Alignment Monitoring System

1. Introduction/Background

This SOW defines the effort required in support of Super – High Energy Replicated Optics (SuperHERO) program which was selected via the National Aeronautics and Space Administration (NASA) Headquarters Research Opportunities in Space and Earth Sciences – 2023 (ROSES-2023) NASA Research Announcement (NRA) Number NNH23ZDA001N, Amendment D.19.

SuperHERO is a high-angular-resolution, balloon-borne, hard-X-ray observatory and is the successor to NASA Marshall Space Flight Center's (MSFC's) successful HERO and HEROES missions. SuperHERO will implement recent advancements in MSFC's replicated nickel shell optics technology to achieve its goal of better than 10 arcsecond-angular-resolution performance in the hard-X-ray (>20 keV) regime. This will allow SuperHERO to achieve its initial science objective of observing the Crab Nebula with unprecedented detail in this bandpass. Flying in the Fall of 2029 from Fort Sumer, New Mexico.

2. Requirements

<u>Alignment Monitoring System (AMS):</u>

- The contractor shall develop and test the AMS for SuperHERO, including the selection of a suitable camera, which will be coordinated with engineers from MSFC.
- The contract shall design a LED pattern to be mounted to the entrance window of each detector. Mechanical and electrical considerations will be coordinated with engineers from MSFC.
- The contractor shall develop the image analysis code to measure the position of the LED pattern in the image. The image analysis code shall be integrated with the SuperHERO flight software. Data products and format shall be coordinated with MSFC.
- In project year two (2), the contractor shall provide eight (8) complete alignment monitoring systems, including camera and LED pattern to be integrated with the payload. In conjunction with MSFC efforts, seven alignment monitoring systems shall be directly integrated with SuperHERO's seven mirror module assemblies and detectors. An additional alignment monitoring system shall be integrated with mass models to be used during hang testing.
- The AMS team shall assist with integration of the system with the mirror module assemblies and the payload at MSFC.
- The contractor shall support payload integration at WFF and at Fort Sumner, New Mexico, and support flight operations.

- The contractor shall support the analysis and publication of the observational results.

3. Deliverables

- The deliverables for this contract are described above as eight complete alignment monitoring systems including the camera and LED pattern integrated with the rest of the payload (supplied separately) including software.
- An additional AMS shall be integrated with mass models to be used during hang testing.
- Support the analysis and publication of the observational results.
- A final report, which can be in PowerPoint, PDF, or Word format, is due within 60 days of the contract completion.

4. Period of Performance

The period of performance for this contract is from the date of the award to December 31, 2030.